

Lab 5: Frontend Aesthetics

Course: Web Development & Databases

Objective

By the end of this lab, students will:

- Understand key principles of frontend aesthetics.
- Learn how to use colors, typography, and spacing effectively.
- Implement a visually appealing web page.

As always, don't just follow the workshop verbatim, tweak the changes and see the difference. Share your findings with the group.

Pre-requisites

- Completion of Lab 4 (Deeper Dive into Frontend Responsive Design).
 - Basic knowledge of HTML, CSS, and JavaScript.
-

Lab Tasks

Task 1: Enhancing Typography and Colors

1. Duplicate `CST150_lab4` folder into `CST150_lab5` folder.
2. Modify `styles.css` to include improved typography and color scheme:

```
body {  
    font-family: 'Poppins', sans-serif;  
    background-color: #f8f9fa;  
    color: #333;  
    line-height: 1.6;  
}  
  
h1, h2, h3 {  
    color: #0056b3;  
}
```

```
a {
  color: #ff6600;
  text-decoration: none;
}

a:hover {
  text-decoration: underline;
}
```

Task 2: Implementing Spacing and Layout Improvements

1. Add proper spacing for readability:

```
.container {
  max-width: 1200px;
  margin: 0 auto;
  padding: 20px;
}

section {
  margin-bottom: 40px;
  padding: 20px;
  background: white;
  border-radius: 8px;
  box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
}
```

Task 3: Using CSS Animations for User Experience

1. Modify `styles.css` to include smooth transitions:

```
button {  
    background: #0056b3;  
    color: white;  
    padding: 10px 20px;  
    border: none;  
    cursor: pointer;  
    transition: background 0.3s ease-in-out;  
}  
  
button:hover {  
    background: #003d82;  
}
```

2. Add a simple animation effect:

```
@keyframes fadeIn {  
    from { opacity: 0; }  
    to { opacity: 1; }  
}  
  
section {  
    animation: fadeIn 1s ease-in-out;  
}
```

Task 4: Testing Aesthetic Improvements

1. Open `index.html` in a browser.
 2. Observe color harmony, spacing, and animations.
 3. Modify colors and fonts to personalise the design.
-

Discussion Questions

1. How do color choices impact user experience?
2. Why is whitespace important in web design?
3. How can subtle animations improve frontend aesthetics?

Bonus Challenge

- Implement a dark mode switch using JavaScript and CSS variables.
 - Add Google Fonts to enhance typography options.
-

Conclusion

In this lab, students explored frontend aesthetics by improving colors, typography, spacing, and animations. Future lessons will focus on frontend frameworks and interactive design elements.

End of Lab 5